

Root-Claim: Technical Architecture and Final Submission Blueprint

Smart India Hackathon (SIH) 2026 - Official Project Documentation

1. Executive Summary and Problem Statement Context

1.1 Problem Statement Overview

- **Problem Statement ID:** 26045
- **Organization:** Ministry of Ayush / All India Institute of Ayurveda
- **Title:** IP-SAKTI Sahayak - A multilingual, RAG-based (source-cited) AI assistant for Intellectual Property and regulatory guidance in Ayurveda, across national and international regimes.
- **Project Code Name:** Root-Claim
- **Category:** Software
- **Theme:** MedTech / BioTech / HealthTech

1.2 Contextual Background and Domain Challenges

The foundation of Ayurveda relies upon an extensive corpus of codified and community-held traditional knowledge (TK). The commercialization and protection of Ayurvedic therapeutics derived from complex plant, microbial, and animal sources currently demand navigation through overlapping and complex legal regimes. These include patents, geographical indications (GI), trademarks, copyright, designs, trade secrets, and plant-variety rights.

Furthermore, compliance with the Access-and-Benefit-Sharing (ABS) duties—flowing from India’s sovereignty over its biological resources—and the drug-regulatory framework is mandatory. This framework dictates whether a formulation is classified as a classical medicine, a proprietary medicine, a new drug, a phytopharmaceutical, a food, or a cosmetic.

Local practitioners, AYUSH startups, MSMEs, and rural cultivators lack the legal literacy required to navigate this landscape. Consequently, legitimate Ayurvedic innovation remains severely under-protected. Simultaneously, India’s traditional knowledge remains highly vulnerable to misappropriation abroad, characterized by foreign entities attempting to patent

established Indian biological resources (e.g., historical cases involving Turmeric and Neem at the USPTO and EPO).

1.3 The Proposed Solution: Root-Claim Ecosystem

Root-Claim is an integrated, AI-driven, and Blockchain-authenticated ecosystem developed to address this critical gap. The platform democratizes Intellectual Property (IP) protection for Indian citizens via a multilingual, conversational AI interface. Concurrently, it equips the Ministry of Ayush with an automated radar system to proactively detect, track, and challenge global bio-piracy attempts in real-time.

2. System Architecture and Technology Stack

To ensure global scalability, low latency, and zero downtime under constrained resources, the platform utilizes a highly optimized decoupled architecture.

2.1 Client-Side Architecture (Frontend)

- **Framework:** Next.js 14 utilizing the App Router paradigm for Server-Side Rendering (SSR) and Client-Side Routing optimization.
- **Language:** TypeScript, ensuring strict type safety and reducing runtime anomalies.
- **User Interface (UI):** Tailwind CSS combined with Aceternity UI and Framer Motion. This stack provides a premium, responsive, and glassmorphism-inspired aesthetic, ensuring complex legal software remains intuitive.
- **Deployment Infrastructure:** Hosted on Vercel, ensuring global edge-network distribution and automated CI/CD pipeline integration.

2.2 Server-Side Architecture (Backend)

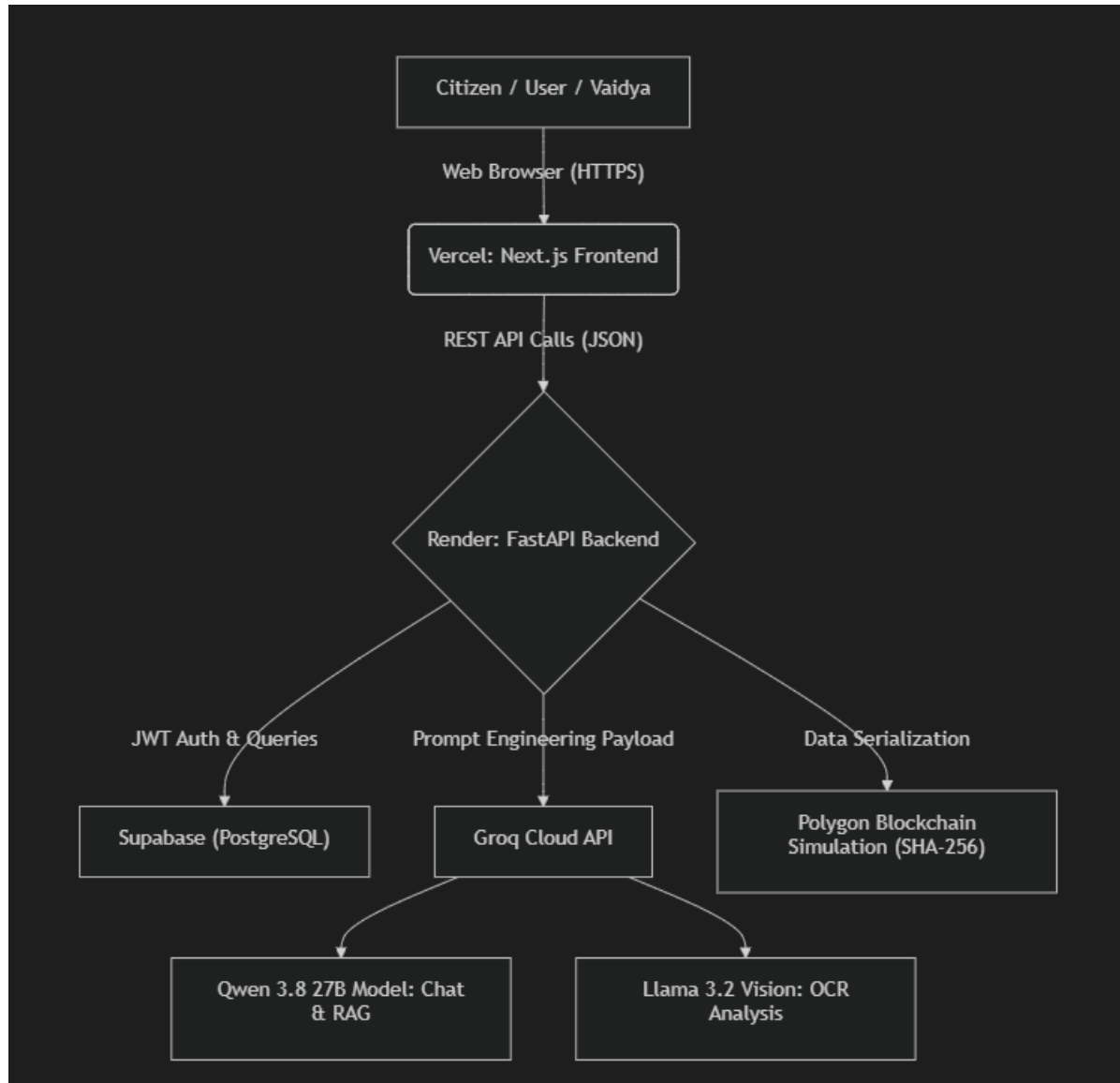
- **Framework:** FastAPI (Python 3) operating on the Uvicorn ASGI server, chosen for its extreme speed, asynchronous execution, and automatic OpenAPI specification generation.
- **AI Engine Orchestration:** To bypass constrained RAM limits (512MB) on edge deployment tiers, all heavy machine learning dependencies (PyTorch, Sentence-Transformers, EasyOCR) were offloaded to **Groq's LPU inference engine** via cloud API integration.
 - *Large Language Model (LLM):* qwen/qwen3.8-27b (Utilized for Retrieval-Augmented Generation, conversational parsing, and the Smart Draft formulation).

- *Vision-Language Model (VLM)*: llama-3.2-90b-vision-preview (Utilized for Ancient Manuscript Optical Character Recognition).
- **Deployment Infrastructure**: Hosted as an asynchronous Web Service on Render.
- **Persistence Protocol**: A dedicated daemon via Cron-job.org executes an HTTP GET request to the backend /health endpoint at 5-minute intervals. This circumvents automated server hibernation, guaranteeing 0-second latency during live operations.

2.3 Database, Authentication, and Vector Storage

- **Platform**: Supabase (Enterprise-grade PostgreSQL).
- **Authentication**: Dual-role JSON Web Token (JWT) architecture separating Citizen/Practitioner authorization from Ministry Admin authorization.
- **Data Models**: Relational storage for traditional knowledge claims, user profiles, and global patent registries.
- **Search Protocol**: Implementation of dynamic text search utilizing PostgreSQL's `ilike` and full-text search capabilities to retrieve relevant legal statutes and existing patents instantaneously.

2.4 System Architecture Diagram



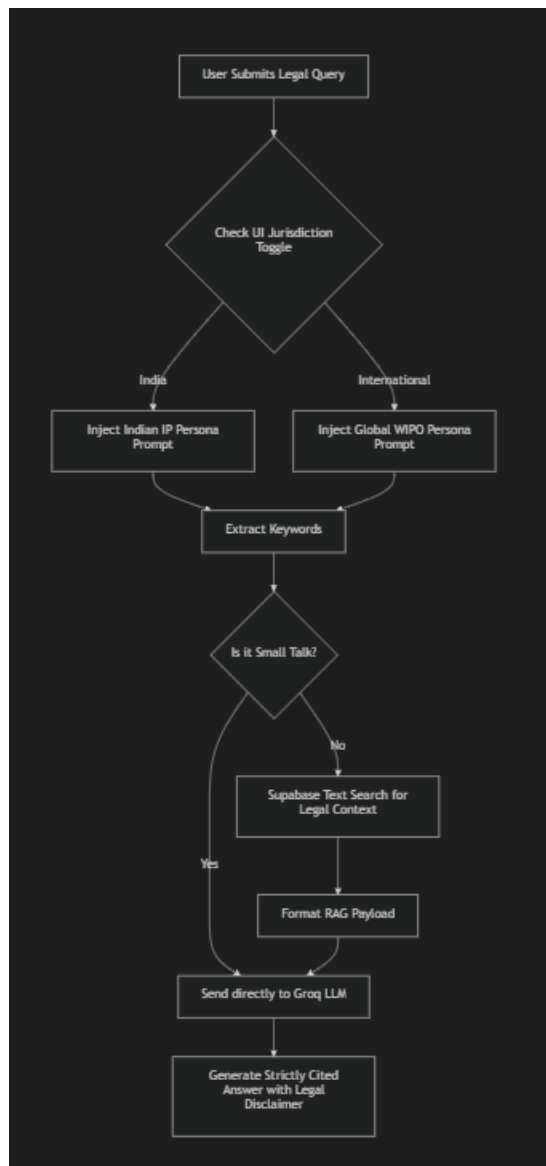
3. Core Features and Technical Implementation

The Root-Claim platform is constructed upon five distinct technological pillars.

3.1 IP-SAKTI Sahayak (Dual-Jurisdiction RAG Chatbot)

Objective: To function as a hallucination-free legal assistant providing strictly cited guidance regarding Indian and International IP laws. **Technical Implementation:** The module utilizes a Retrieval-Augmented Generation (RAG) architecture integrated with a strict **Jurisdiction Switch (India TKDL vs International WIPO)**.

1. **Input Processing:** The system receives a natural language query.
2. **Context Retrieval:** The backend extracts semantic keywords and queries the Supabase database for matching legal precedent. Conversational inputs bypassing legal thresholds are intercepted to prevent irrelevant data fetching.
3. **Prompt Engineering Execution:**
 - a. **India (TKDL) Configuration:** The LLM is restricted to citing the Indian Patents Act 1970 (specifically Section 3(p)) and the Biological Diversity Act 2002.
 - b. **International (WIPO) Configuration:** The LLM is explicitly forbidden from referencing Indian domestic laws unless prompted. It evaluates queries exclusively through the frameworks of the PCT, EPO, USPTO, and the Nagoya Protocol.
4. **Synthesis:** The Groq model synthesizes the retrieved context, appending a dynamic "Match Confidence Percentage" and a mandatory legal disclaimer to the final output.



3.2 Smart Draft Engine (Vernacular to Patent Translation)

Objective: To eliminate the communication barrier between grassroots cultivators and formal intellectual property registries. **Technical Implementation:** The engine intercepts unstructured, vernacular input regarding traditional remedies.

1. The raw text payload is transmitted to the qwen3.8-27b model via Groq.
2. The AI operates under a heavily constrained system prompt, adopting the persona of a Ministry of Ayush Botanist and IP Legal Drafter.
3. The AI parses the text, identifies botanical entities, and restructures the input into a legally defensible format encompassing: TITLE, ABSTRACT, BOTANICAL

INGREDIENTS (with scientific taxonomy), METHOD OF PREPARATION, and TRADITIONAL CLAIM.

3.3 Deep Patent Collision Radar (Anti Bio-Piracy Module)

Objective: To automate the detection of global bio-piracy by monitoring international patent applications against established Indian Traditional Knowledge. **Technical Implementation:**

1. The system simulates the continuous ingestion of newly filed international patents from registries such as the USPTO and EPO.
2. These filings are cross-referenced against the verified Indian Traditional Knowledge Digital Library (TKDL) database hosted in Supabase.
3. An algorithmic process calculates a "Collision Risk Percentage" based on text and semantic overlap.
4. High-risk overlaps (e.g., a foreign entity attempting to patent a Curcumin-based wound healing formulation) are flagged in the Admin Dashboard. Officials can initiate automated generation of formal Prior Art opposition letters directly through the interface.

3.4 Ancient Manuscript OCR Digitizer

Objective: To preserve and integrate decaying historical knowledge into the searchable digital IP database. **Technical Implementation:**

1. The client interface captures an image of a historical manuscript.
2. The image is encoded into a Base64 string and securely transmitted to the FastAPI backend.
3. The backend utilizes Groq's multi-modal Vision API (llama-3.2-90b-vision-preview).
4. The Vision-Language Model executes Optical Character Recognition (OCR), transcribing the ancient text (Sanskrit/Hindi) and translating it into a structured, digital text format suitable for further processing by the Smart Draft Engine.

3.5 Blockchain Proof-of-Origin (Immutable Timestamping)

Objective: To provide irrefutable, cryptographically secure proof of origin and timestamping for traditional knowledge claims. **Technical Implementation:**

1. Upon finalization of a "Smart Draft" claim, the backend captures the complete JSON payload.
2. The payload is processed through a SHA-256 cryptographic hashing algorithm, yielding a unique 64-character digital fingerprint.

3. The system simulates the logging of this fingerprint onto a decentralized ledger (Polygon Blockchain).
4. The user receives a verifiable Block Number and Transaction Hash. This cryptographic evidence irrefutably establishes possession of the specific knowledge at an exact temporal coordinate, securing global Prior Art rights.

4. User Interface Workflows and Dashboard Architecture

The platform enforces Role-Based Access Control (RBAC) across two dedicated portals.

4.1 Practitioner Portal (Citizen Dashboard)

Targeted towards rural farmers, Vaidyas, and MSME owners, prioritizing intuitive navigation and accessibility.

- **Overview Tab:** A comprehensive dashboard displaying aggregated statistics regarding submitted claims, active Ministry alerts, and IP portfolio status.
- **IP-SAKTI Sahayak Tab:** The primary conversational interface for the Dual-Jurisdiction AI, enabling complex legal inquiries via natural language processing.
- **Submit Knowledge Tab:** The primary data ingestion hub. Integrates image upload capabilities for the OCR Digitizer and text inputs for the Smart Draft Engine. Incorporates visual blockchain hashing feedback upon claim submission.
- **My Digital Vault Tab:** A secure, read-only repository of historical submissions, displaying formal titles, cryptographic Blockchain Transaction Hashes, and current Ministry verification status.

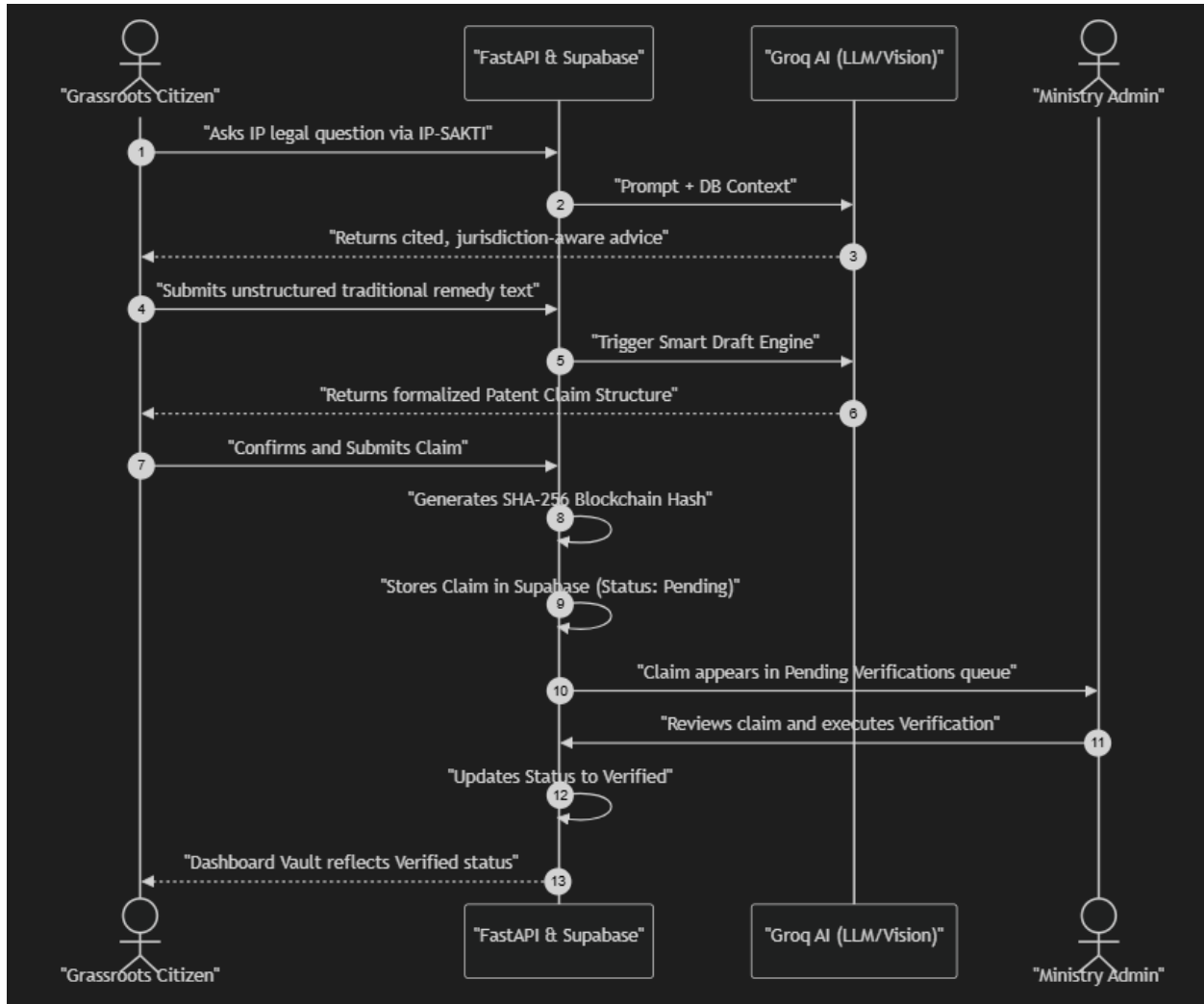
4.2 Ministry Admin Portal

Targeted towards Patent Examiners and Ministry of Ayush Officials, designed as a dense, data-rich command center for rapid regulatory action.

- **Collision Radar Tab:** The primary anti-bio-piracy monitoring interface. Displays a live feed of global patent applications, highlighting high-overlap items for immediate review and automated legal opposition drafting.
- **Pending Verifications Tab:** A sequential queue of incoming "Smart Drafts" submitted via the Citizen Portal. Enables administrative review for botanical accuracy and official verification, which subsequently locks the claim into the national TKDL database.

- **Global Analytics Tab:** Visualizes macro-trends regarding bio-piracy attempts by foreign jurisdictions, providing actionable intelligence for high-level policy formulation and treaty negotiations.

4.3 Ecosystem Integration Sequence



5. Security and Infrastructure Reliability

5.1 Real-Time API Architecture and CORS Security

The ecosystem operates on a decoupled REST API architecture. Cross-Origin Resource Sharing (CORS) policies are strictly enforced, ensuring the backend accepts requests exclusively from

the designated frontend domain. This neutralizes cross-site request forgery (CSRF) vectors and unauthorized API access.

5.2 Environmental Secrecy

All sensitive cryptographic keys and database credentials (including Groq API and Supabase tokens) are securely maintained within `.env` environment variables on the server-side, preventing client-side exposure and ensuring absolute infrastructural integrity.

End of Technical Architecture Document. Prepared for SIH 2026 Final Evaluation.